8.4 Spiralling onto a Circle

WORKSHEET

The question

(i) Show that under the changes of variable $x = r\cos\theta$ and $y = r\sin\theta$, where r is a function of θ with $r > 0$, the differential equation

$$(y + x)\frac{\mathrm{d}y}{\mathrm{d}x} = y - x$$

becomes

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} + r = 0.$$

Sketch a solution in the x-y plane.

(ii) Show that the solutions of

$$\left(y + x - x(x^2 + y^2)\right)\frac{\mathrm{d}y}{\mathrm{d}x} = y - x - y(x^2 + y^2)$$

can be written in the form

$$r^2 = \frac{1}{1 + A\mathrm{e}^{2\theta}}$$

and sketch the different forms of solution that arise according to the value of A.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself Both equations contain the combinations $x\,\mathrm{d}x + y\,\mathrm{d}y$ and $x\,\mathrm{d}y - y\,\mathrm{d}x$ in disguise. What are these in polar coordinates?

STAGE 1 Part (i) in polar coordinates

Ask yourself Check that rearrangement by expanding. Now use the two identities above.

STAGE 2 Part (ii)

Pause and try The new equation is the old one plus extra terms involving $x^2 + y^2 = r^2$. Rewrite it in the same way and find $\frac{\mathrm{d}r}{\mathrm{d}\theta}$.

Ask yourself This is not linear in r, but it becomes linear in $u = r^{-2}$. What equation does u satisfy?

Try this next

1. **The spiral.** For the solution $r = e^{-\theta}$ of part (i), show that the curve crosses every ray from the origin at the same angle, $45°$. (This is why logarithmic spirals are called *equiangular*.)
2. **Polar identities.** Prove that $x\,\mathrm{d}x + y\,\mathrm{d}y = r\,\mathrm{d}r$ and $x\,\mathrm{d}y - y\,\mathrm{d}x = r^2\,\mathrm{d}\theta$.
3. **A system in time.** Consider the system $\dot{r} = r(1 - r^2)$, $\dot{\theta} = 1$. Show that every solution with $r(0) > 0$ approaches the unit circle as $t \to \infty$. Write the system in terms of x and y.
4. **Two cycles.** Find a system of the form $\dot{r} = f(r)$, $\dot{\theta} = 1$ with two limit cycles, at $r = 1$ and $r = 2$, one attracting and one repelling.

Further reading

- **Steven Strogatz, *Nonlinear Dynamics and Chaos*** (CRC Press). Chapter 7 is about limit cycles and the Poincaré–Bendixson theorem.
- **Steven Strogatz, *Sync*** (Hyperion). A popular book about rhythms and synchronisation in nature.

- **James Gleick, *Chaos: Making a New Science*** (Penguin). The story of Lorenz and the discovery of chaos.

The full guided solution, and what the question leads to at university, is in Section 8.4:
https://keithmcnulty.github.io/step_book_mcnulty/ch08/limit-cycle.html